Worksheet WK 6-9 E
U-Substitution and Inverse Trig Functions
AP Calculus

Name_____

Evaluate each integral.

1.) $\int \frac{dx}{9x^2 + 16}$

2.) $\int \sin x\,(\cos x + 3)^{\frac{3}{4}}\,dx$

3.) $\int \frac{dx}{\sqrt{4 - 9x^2}}$

4.) $\int x^5\, 8^{7x^6}\,dx$

5.) $\int \frac{\tan(\tan x)}{\cos^2 x}\,dx$

6.) $\int \sin(5x)\cos(5x)\,dx$

7.) $\int \frac{dx}{x\sqrt{x^2 - 25}}$

8.) $\int \frac{e^{-x}}{1 + e^{-2x}}\,dx$

9.) $\int \frac{dx}{11 + x^2}$

10.) $\int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}}$

11.) $\int_{-1}^{1} \frac{dx}{1+x^2}$

12.) $\int_{-1}^{0} \frac{4\,dx}{1+x^2}$

13.) $\int_0^{\frac{\sqrt{2}}{2}} \frac{x\,dx}{\sqrt{1-x^4}}$

14.) $\int_0^{2} \frac{dx}{1+(x-1)^2}$

15.) $\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \frac{2\cos x\ dx}{1+\sin^2 x}$

16.) $\int_0^{\sqrt{2}} \frac{4x\,dx}{\sqrt{4-x^4}}$